Introduction: Stochastic partial differential equations

Some motivating examples:

Example 1: Dean-Kawasaki equation (Dean, 1996, Kawasaki, 1998)
Consider independent Brownian motions $B^1, B^2, B^3, \dots$ and $f \in C_b^2(\mathbb{R})$, then the following strong Law of Large Numbers (LLN) holds:

$$\eta_t^N(f) := \frac{1}{N}\sum_{i=1}^N f(B_t^i + x_0) \to \mathbb{E}\left(f(B_t + x_0)\right) \text{ as } N \to \infty,$$

that is

$$\underbrace{\eta_t^N(dx)}_{random} \to \underbrace{p(t,x)\,dx}_{det.} \text{ as } N \to \infty.$$

where $p(t,x)$ solves the **heat equation**

$$\begin{cases} \partial_t p(t,x) &= \frac{1}{2}\partial_{xx} p(t,x), \quad t \in [0,T], x \in \mathbb{R} \\ p(0,x)dx &= \delta_{x_0}(dx) \end{cases} \tag{1.1}$$

as a special case of the **Fokker-Planck equation** for the time evolution of the density of a diffusion process. In this particular case, the solution can be explicitly given as the density of the normal distribution $\mathcal{N}(x_0, t)$:

$$p_t(x_0, x) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{(x-x_0)^2}{2t}\right).$$

The above convergence statement poses a couple of interesting and currently still open mathematical problems, that we will briefly discuss, as far as we can get without the theory of stochastic partial differential equations yet.

For notational convenience let $\eta_t(dx) := p(t,x)\,dx$ denote the limiting Gaussian distribution. Then

$$\mathbb{E}\left(\eta_t^N(f)\right) = \mathbb{E}\left(f(B_t + x_0)\right) = \int_{\mathbb{R}} f(x)\eta_t(dx) = \eta_t(f)$$

so that we obtain the approximation error

$$\mathbb{E}\left((\eta_t^N(f) - \eta_t(f))^2\right) = \frac{1}{N}\mathrm{Var}\left(f(B_t + x_0)\right) = O\left(N^{-1}\right).$$

Fluctuations of $\sqrt{N}(\eta_t^N(f) - \eta_t(f))$

Applying Itô's formula yields

$$
\begin{aligned}
d\eta_t^N(f) &= d\left(\frac{1}{N}\sum_{i=1}^N f(B_t^i + x_0)\right) \\
&= \underbrace{\frac{1}{N}\sum_{i=1}^N \partial_x f(B_t^i + x_0) dB_t^i}_{=:dM_t^N(f)} + \underbrace{\frac{1}{N}\sum_{i=1}^N \frac{1}{2}\partial_{xx} f(B_t^i + x_0)}_{=\eta_t^N(\frac{1}{2}\partial_{xx} f)}\, dt \\
&= dM_t^N(\partial_x f) + \eta_t^N(\frac{1}{2}\partial_{xx} f)\, dt\,.
\end{aligned} \tag{1.2}
$$

Hence, if $\psi_t^N(f) := \sqrt{N}(\eta_t^N(f) - \eta_t(f))$ denotes the standardized fluctuations, (1.2) and (1.1) imply the following stochastic different equation

$$
d\psi_t^N(f) = \sqrt{N} dM_t^N(\partial_x f) + \psi_t^N(\frac{1}{2}\partial_{xx} f)\, dt\,. \tag{1.3}
$$

The rescaled (local) martingale $\sqrt{N} M_t^N(\partial_x f)$ has quadratic variation

$$
\begin{aligned}
\langle \sqrt{N} M_\cdot^N(\partial_x f)\rangle_t &= \frac{1}{N}\langle \sum_{i=1}^N \int_0^\cdot \partial_x f(B_s^i + x_0) dB_s^i\rangle_t \\
&= \frac{1}{N}\sum_{i=1}^N \langle \int_0^\cdot \partial_x f(B_s^i + x_0) dB_s^i\rangle_t \\
&= \frac{1}{N}\sum_{i=1}^N \int_0^t \left(\partial_x f(B_s^i + x_0)\right)^2\, ds \\
&= \int_0^t \eta_s^N((\partial_x f)^2)\, ds \to \int_0^t \eta_s((\partial_x f)^2)\, ds\,, N \to \infty\,,
\end{aligned}
$$

again by the strong LLN. The martingale central limit theorem now implies that $\sqrt{N} M_t^N(\partial_x f)$ converges in distribution to a Brownian motion $W_t(\partial_x f)$ with variance

$$
\int_0^t \eta_s((\partial_x f)^2\, ds
$$

which implies that $\psi_t^N(\partial_x f)$ also weakly converges to a stochastic process that can be characterized as a solution of the stochastic differential equation

$$
d\psi_t(f) = dW_t(\partial_x f) + \psi_t(\frac{1}{2}\partial_{xx} f)\, dt\,. \tag{1.4}
$$

Our next step will be to obtain from (1.4) an explicit equation in ψ_t that does no longer involve the test function f. This can be achieved if we can realize $\psi_t(f)$ as an integral

$$
\psi_t(f) = \int_{\mathbb{R}} f(x)\psi_t(x)\, dx
$$

w.r.t. a signed density, simply a function, in C^2, since integration by parts then implies

$$
\psi_t(\frac{1}{2}\partial_{xx} f) = \int_{\mathbb{R}} f(x)\frac{1}{2}\partial_{xx}\psi_t(x)\, dx.
$$

Also suppose that we can identify the Brownian motion $W_t(\partial_x f)$ as a stochastic integral

$$-\int_0^t \int_{\mathbb{R}} \partial_x f(x) \sqrt{\eta_t(x)}\, dW(t,x) \tag{1.5}$$

w.r.t. a suitable space-time stochastic process $W(t,x)$, then integration by parts

$$-\int_0^t \int_{\mathbb{R}} \partial_x f(x) \sqrt{\eta_t(x)}\, dW(t,x) = \int_0^t \int_{\mathbb{R}} f(x) \partial_x \left(\sqrt{\eta_t(x)} dW(t,x) \right)$$

in the above equation yields the following explicit **stochastic partial differential equation** (SPDE)

$$\partial_t \psi_t = \frac{1}{2} \partial_{xx} \psi_t + \partial_x \left(\sqrt{\eta_t(x)} dW(t,x) \right) \tag{1.6}$$

which is called the `Dean-Kawasaki SPDE`.

To obtain the representation (1.5), we rewrite the Brownian motion $W_t(\partial_x f)$, using Parseval's identity, as

$$\eta_t((\partial_x f)^2) = \int_{\mathbb{R}} (\partial_x f(x))^2 \eta_t(x)\, dx = \sum_{k=1}^{\infty} \langle \partial_x f \sqrt{\eta_t}, e_k \rangle^2$$

for any given complete orthonormal system (CONS) $(e_k)_{k \geq 1}$ of the Hilbert space $L^2(\mathbb{R})$. Taking a family of independent Brownian motions $(\hat{B}_t^k)_{t \geq 0}$, we can apply Wiener-Itô isometry to obtain

$$\int_0^t \langle \partial_x f \sqrt{\eta}_s, e_k \rangle^2\, ds = \hat{\mathbb{E}} \left(\int_0^t \int_{\mathbb{R}} \partial_x f(x) \sqrt{\eta}_s(x) e_k(x)\, dx d\hat{B}_s^k \right)^2,$$

where $\hat{\mathbb{E}}$ denotes expectation w.r.t. the Brownian motions $\hat{B}^k$, $k = 1, 2, 3, \ldots$. Summing up and using independence of the stochastic integrals $\int_0^t \int_{\mathbb{R}} f'(x) \sqrt{\eta}_t(x) d\hat{B}_t^k$, $k = 1, 2, 3, \ldots$, we obtain that

$$\begin{aligned}
\eta_t((\partial_x f)^2) &= \int_{\mathbb{R}} (\partial_x f(x))^2 \eta_t(x)\, dx \\
&= \sum_{k=1}^{\infty} \hat{\mathbb{E}} \Big(\Big(\int_0^t \int_{\mathbb{R}} \partial_x f(x) \sqrt{\eta}_s(x) e_k(x)\, dx d\hat{B}_s^k \Big)^2 \Big) \\
&= \hat{\mathbb{E}} \Big(\Big(\sum_{k=1}^{\infty} \int_0^t \int_{\mathbb{R}} \partial_x f(x) \sqrt{\eta}_s(x) e_k(x)\, dx d\hat{B}_s^k \Big)^2 \Big) \\
&= \hat{\mathbb{E}} \Big(\Big(\int_0^t \int_{\mathbb{R}} \partial_x f(x) \sqrt{\eta}_s(x) \underbrace{\big(\sum_{k=1}^{\infty} e_k(x)\, dx d\hat{B}_s^k \big)}_{:= dW(s,x)} \Big)^2 \Big).
\end{aligned}$$

$dW(t,x)$ is called **space-time white noise**, since for any bounded disjoint rectangles $]s_i, t_i] \times [x_i, y_i]$, the integrals

$$\begin{aligned}
W(]s_i, t_i] \times [x_i, y_i]) &:= \int_0^T 1_{]s_i, t_i] \times [x_i, y_i]}(t,x) dW(t,x) \\
&= \sum_{k=1} \int_0^T 1_{]s_i, t_i]}(t) \int_{\mathbb{R}} 1_{[x_i, y_i]}(x) e_k(x)\, dx\, d\hat{B}_t^k
\end{aligned}$$

are independent Gaussian random variables. (Joint) Gaussianity is obvious, independence follows from

$$
\begin{aligned}
&\hat{\mathbb{E}}\left(W(]s_i,t_i]\times[x_i,y_i])W(]s_j,t_j]\times[x_j,y_j])\right)\\
&\quad=\sum_{k,l=1}\hat{\mathbb{E}}\left(\int_0^T 1_{]s_i,t_i]}(t)\int_{\mathbb{R}}1_{[x_i,y_i]}(x)e_k(x)\,dx\,d\hat{B}_t^k\int_0^T 1_{]s_i,t_i]}(t)\int_{\mathbb{R}}1_{[x_j,y_j]}(x)e_l(x)\,dx\,d\hat{B}_t^l\right)\\
&\quad=\sum_{k=1}\hat{\mathbb{E}}\left(\int_0^T 1_{]s_i,t_i]}(t)\int_{\mathbb{R}}1_{[x_i,y_i]}(x)e_k(x)\,dx\,d\hat{B}_t^k\int_0^T 1_{]s_i,t_i]}(t)\int_{\mathbb{R}}1_{[x_j,y_j]}(x)e_k(x)\,dx\,d\hat{B}_t^k\right)\\
&\quad=\sum_{k=1}\int_0^T 1_{]s_i,t_i]}(t)1_{]s_j,t_j]}(t)\int_{\mathbb{R}}1_{[x_i,y_i]}(x)e_k(x)\,dx\int_{\mathbb{R}}1_{[x_j,y_j]}(x)e_k(x)\,dx\,dt\\
&\quad=\sum_{k=1}\int_0^T 1_{]s_i,t_i]}(t)1_{]s_j,t_j]}(t)\int_{\mathbb{R}}1_{[x_i,y_i]}(x)1_{[x_j,y_j]}(x)\,dx\,dt=0,\, i\neq j,
\end{aligned}
$$

where in the second but last line we used the Wiener-Itô isometry and in the last line the Parseval identity.

Remark The solution theory of the SPDE (1.6) is largely open, since its precise mathematical formulation is unclear. Can the asymptotic fluctuations $\psi_t(f)$ really be represented as the integral of f w.r.t. a (signed) density, let alone sufficiently differentiable density? The Appendix at the end of this introduction, contains some additional analysis of the asymptotic behaviour of the (signed) densities $\psi_t^N(x)$ if they exist. This requires in particular, that the initial condition of the Brownian motions have a density. The analysis on the signed density yields a limiting SPDE that apparently looks different to (1.6), indicating that some assumptions made in the above reasoning are not easy to justify rigorously.

Example 2: Zakai- and Kushner Stratonovich equation

Consider the filtering problem of a signal process (S) and observations (O), given in terms of solutions to the following stochastic differential equations

$$
\begin{aligned}
&(S) \qquad dX_t = B(t,X_t)\,dt + C(t,X_t)\,dW_t\\
&(O) \qquad dY_t = H(t,X_t)\,dt + \Gamma(t)\,dV_t
\end{aligned}
\tag{1.7}
$$

with standard assumptions on the coefficients ensuring strong well-posedness. Also suppose that the **measurement covariance** $R_t := \Gamma(t)\Gamma^T(t)$ is strictly positive definite.

Let $\mathcal{Y}_t := \sigma\left(Y_s \mid s\in[0,t]\right)$ be the σ-algebra generated by the observations up to time t. The stochastic filtering problem now consists of computing resp. approximating the conditional distribution

$$
\eta_t(A) = \mathbb{P}\left(X_t\in A \mid \mathcal{Y}_t\right) \tag{1.8}
$$

of the signal process given the information $\mathcal{Y}_t$ provided by the observations up to time t. It turns out that η_t can be represented as

$$
\eta_t(A) := \frac{\psi_t(A)}{\psi_t(\mathbb{R}^d)} = \frac{\mathbb{E}\left(1_A(X_t)Z_t\right)}{\mathbb{E}\left(Z_t\right)}, \quad A\in\mathcal{B}(\mathbb{R}^d), \tag{1.9}
$$

where

$$Z_t = \exp\left(\int_0^t \langle R_s^{-1} H(s,X_s), dY_s\rangle - \frac{1}{2}\int_0^t \langle R_s^{-1} H(s,X_s), H(s,X_s)\rangle\, ds\right). \tag{1.10}$$

The Zakai equation

Itô calculus applied to $f(X_t)Z_t$ for $f \in C_b^2$ implies that

$$df(X_t)Z_t = L_t f(X_t) Z_t\, dt + f(X_t) Z_t R_t^{-1} H(t,X_t) dY_t + dM_t(f) \tag{1.11}$$

for the (local) martingale $M_t(f) = \int_0^t \nabla f(X_s)\, C(s,X_s)\, dX_s$. Here,

$$L_t f(x) = \frac{1}{2}\sum_{i,j=1}^d a_{ij}(t,x)\frac{\partial^2 f}{\partial x_i \partial x_j}(x) + \sum_{i=1}^d b_i(t,x)\frac{\partial f}{\partial x_i}(x)\,, \tag{1.12}$$

with $B(t,x) = [b_1(t,x),\dots,b_d(t,x)]$ and

$$a_{ij}(t,x) = (CC^T)_{ij}\ (t,x) \qquad \text{"diffusion matrix"}$$

denotes the generator of the signal process (S). The covariation between $f(X_t)$ and Z_t vanishes, since W and V are supposed to be independent Brownian motions. Taking expectation (only w.r.t. the Brownian motion driving (S)), we arrive at the integral equation

$$\psi_t(f) = \psi_0(f) + \int_0^t \psi_s(L_s f)\, ds + \int_0^t \psi_s(f R_s^{-1} H(t,\cdot)) dY_s. \tag{1.13}$$

Assuming that $\psi_t(dx) = \psi_t(x)\, dx$ admits a density, we can integrate by parts and obtain

$$\int f\psi_t\, dx = \int f\, d\eta_0 + \int f\left(\int_0^t L_s^*\psi_s\, ds\right) dx + \int f\left(\int_0^t \psi_s R_s^{-1} H(s,\cdot) dY_s\right) dx\,.$$

where

$$L_t^* f(x) := \frac{1}{2}\sum_{ij}\frac{\partial^2(a_{ij}(t,\cdot)g)}{\partial x_i \partial x_j} - \sum_i \frac{\partial(b_i(t,\cdot)g)}{\partial x_i} \tag{1.14}$$

denotes the formal adjoint of the generator L_t. Since the last equality holds for all $f \in C^2(\mathbb{R}^d)$ with compact support, this implies that

$$\psi_t(x) = \eta_0(x) + \int_0^t L_s^*\psi_s(x)\, ds + \int_0^t \psi_s(x) R_s^{-1} H(s,x) dY_s\,.$$

In other words: the unnormalized conditional density satisfies the SPDE

$$\partial_t \psi_t(x) = L_t^*\psi_t(x)\, dt + \psi_t(x) R_t^{-1} H(t,x) dY_t \tag{1.15}$$

which is called the **Zakai equation**.

Kushner-Stratonovich equation

Likewise, $\eta_t(f) = \mathbb{E}\left(f(X_t) \mid \mathcal{Y}_t\right)$ satisfies the **Kushner-Stratonovich** equation

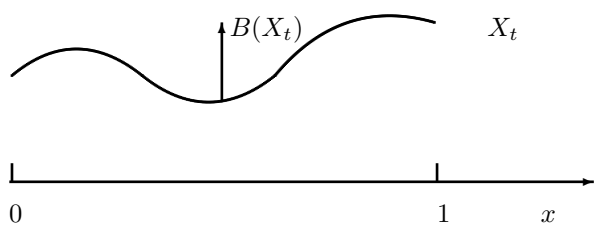
$$d\eta_t(f) = \eta_t(L_t f)\, dt + \langle \eta_t(f R_t^{-1} H(t,\cdot)) - \eta_t(f)\eta_t(R_t^{-1} H(t,\cdot)), \underbrace{dY_t - \eta_t(H(t,\cdot))\, dt}_{=:dI_t}\rangle \tag{1.16}$$

where

$$dI_t = dY_t - \eta_t(H(t,\cdot))\, dt$$

is called the **innovation** term.

Example 3: random motion of strings (Funaki, 1983)



where

- $X_t : [0,1] \to \mathbb{R}$ is a (random) function on $[0,1]$ ('string') and $X_t(x)$ denotes its value at position $x \in [0,1]$ (random variable (!))
- $B : \mathbb{R} \to \mathbb{R}$ models forcing term, $B(X_t(x))$ models direction of some exterior force acting on the string at position x
- time evolution of the moving string:

$$\partial_t X_t = \nu \underbrace{\partial^2_{xx} X_t}_{\text{diffusion}} + B(X_t) + \sigma \underbrace{\partial_t W_t}_{\text{noise}}$$

subject to boundary conditions:

- Neumann bdy cond: $\partial_x X_t(0) = \partial_x X_t(1) = 0$
- Dirichlet bdy cond: $X_t(0) = X_t(1) = 0$

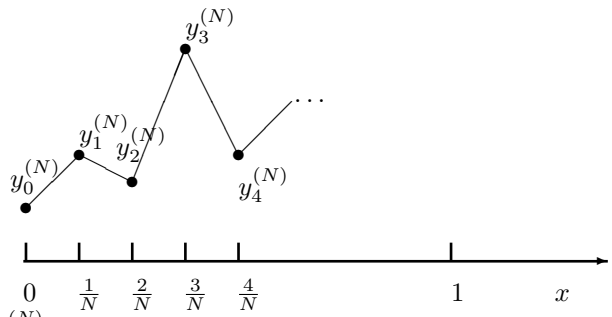
Here, W_t denotes a random function such that for all $x \in [0,1]$

$$t \mapsto W_t(x)$$

is a standard Brownian motion and for $x \neq y$ the Brownian motions $W_\cdot(x)$ and $W_\cdot(y)$ should be for simplicity independent. ('space-time white noise')

What is a rigorous mathematical description of this?

spatial discretization given $N \in \mathbb{N}$



The nodes $y_k^{(N)}(t)$, $k = 0, 1, \ldots, N$, are solutions of the following system of stochastic differential equations:

$$dy_k^{(N)}(t) = \nu \underbrace{\frac{1}{(\Delta x)^2}\left(y_{k+1}^{(N)}(t) - 2y_k^{(N)}(t) + y_{k-1}^{(N)}(t)\right)}_{\text{2nd order finite diff.}} dt$$
$$+ B(y_k^{(N)}(t))\, dt + \frac{1}{\sqrt{\Delta x}} dW_k(t),\ k = 0, 1, \ldots, N.$$

with $\Delta x = \frac{1}{N}$ ('spatial distance').

Why spatial scaling $\frac{1}{\sqrt{\Delta x}}$ in front of the Brownian motions ?

associated polygon:

$$X^{(N)}(t,x) := \frac{x_{k+1}^{(N)} - x}{x_{k+1}^{(N)} - x_k^{(N)}} y_k^{(N)}(t) + \frac{x - x_k^{(N)}}{x_{k+1}^{(N)} - x_k^{(N)}} y_{k+1}^{(N)}(t) \text{ if } x \in [x_k^{(N)}, x_{k+1}^{(N)}]$$
$$\in C([0,1]) \subset L^2([0,1])$$

with $x_k^{(N)} := \frac{k}{N}$.

Functional central limit theorem

Let $P^{(N)} := P \circ (X^{(N)})^{-1}$ on $C([0,1]; L^2([0,1]))$. Then $P^{(N)} \Rightarrow P^{(\infty)}$ weakly on $C([0,1]; L^2([0,1]))$ where the canonical process under $P^{(\infty)}$ solves the stochastic partial differential equation (spde)

$$dX_t = \left(\nu \partial_{xx}^2 X_t + B(X_t)\right) dt + dW_t \quad \in L^2([0,1]) \tag{1.17}$$

where W_t denotes the 'Brownian motion' on $L^2([0,1])$.

Formally: $\forall\ s, t \in [0, \infty)$, $\forall\ x, y \in [0,1]$

$$\mathbb{E}\left(W_s(x) W_t(y)\right) = s \wedge t\ \delta_0(x - y)$$

Alternative notation for the spde (1.17): Using $u(t,x) := X_t(x)$ and $W(t,x) = W_t(x)$, we can formally write

$$\partial_t u(t,x) = \nu \partial_{xx}^2 u(t,x) + B(u(t,x)) + \partial_t W(t,x) \quad \in L^2([0,1]) \tag{1.18}$$

Of course, the time-derivative of the Wiener process does not exist in the classical sense and (1.18) has to be understood as usual in the sense of the associated integral equation.

Example 4: stochastic reaction-diffusion equations

A large class of widely used stochastic models for spatially distributed random evolutions are given by stochastic reaction-diffusion equations, e.g.

$$\begin{cases} \partial_t v(t,x) &= \nu \partial^2_{xx} v(t,x) + b f(v(t,x)) + g(v(t,x)) \partial_t W(t,x) \\ v(0,x) &= v_0(x) \end{cases} \tag{1.19}$$

where

- $x \in D \subset \mathbb{R}^d$
- $\lim_{|v| \to \infty} f(v) \cdot v = -\infty$, i.e., 'coercive'
- $(f(u) - f(v))(u - v) \leq |u - v|^2$, i.e., f one-sided Lipschitz
  - $f(v) = v(1-v)(v-a)$ - Nagumo
  - $f(v) = v(1-v)$ - Fisher resp. KPP (Kolm.-Petrovsky-Piskounov)
  - $f(v,w) = [v(1-v)(v-a) - w, \epsilon(v - \gamma w)]$ - FitzHugh-Nagumo
  - $f(v,w) = [a - (b+1)v + v^2 w, w = bv - v^2 w]$ - Brusselator

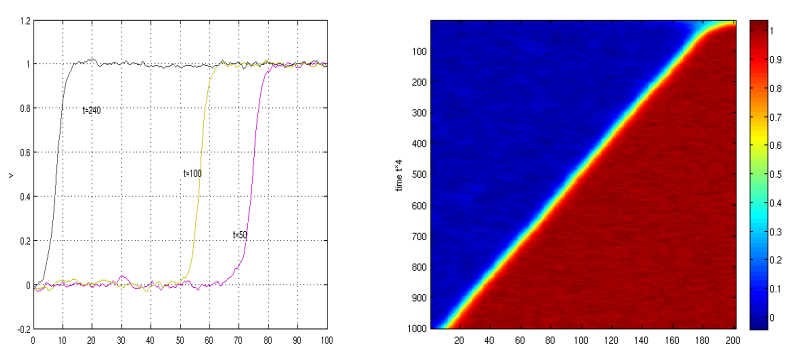
Motivation mathematical models for spatially extended random evolutions used in physical chemistry, material sciences, optics, biophysics, population biology

Dynamical features of (1.19) **of interest in the applications**

- bifurcations, i.e., qualitative different dynamical behaviour depending on the chosen set of parameters (e.g. excitable, oscillatory, ...)
- metastable and in general transient macroscopic structures (pulses, wave fronts, solitons, ...)
- stochastic resonance, i.e., microscopic stochastic fluctuations can accumulate leading to change points on macroscopic scales (e.g. in climate models)
- superposition and interaction of physical/dynamical processes on various time scales and various spatial scales

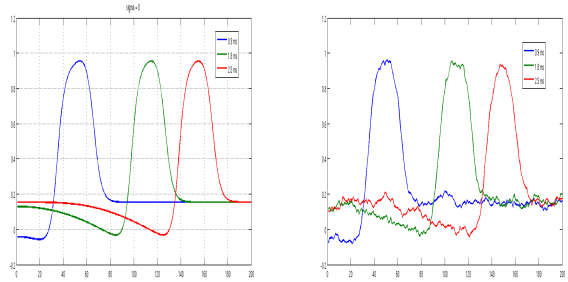
1D-stochastic Nagumo equation

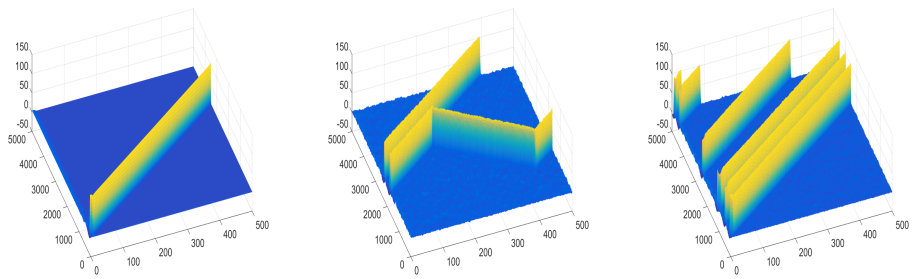
exhibits travelling waves moving with random speed



1D-stochastic FitzHugh-Nagumo system

exhibits travelling action potentials (AP's) moving with random speed. Other features include random spiking, multiple spikes, annihilation of spikes, ...





2D-Stochastic FitzHugh-Nagumo systems

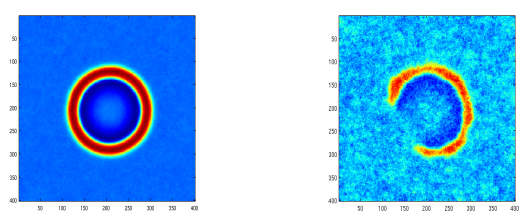
model for intracellular signalling pathways

$$\partial_t v(t,x,y) = \partial^2_{xx} v(t,x,y) + \partial^2_{yy} v(t,x,y) + (v(1-v)(v-a))(t,x,y) - w(t,x,y) + I(t,x,y) + \sigma \partial_t W(t,x,y)$$

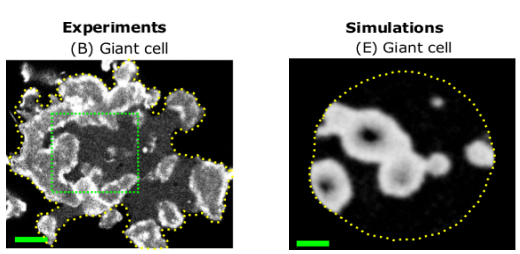
$$\partial_t w(t,x,y) = \epsilon(v - \gamma w)(t,x,y)$$

on the domain $[0,L] \times [0,L] \subset \mathbb{R}^2$

exhibits various spatial patterns



comparison with real experiments



Wave dynamics in the actin cytoskeleton (in *Dictyostelium discoideum*, see [5])